Chapter 12

Quantifying Uncertainty

Russell & Norvig – AI: A Modern Approach, 4th Edition

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Spring 2026

Agenda

Part 1

Why Uncertainty?

- Limits of logic
- Degree of belief
- Rational decisions

Part 2

The Language of Probability

- Axioms
- Bayes' rule
- Conditional probability

Part 3

Independence & Naïve Bayes

- Joint distributions
- Conditional independence
- Naïve Bayes model

PART 1

Why Uncertainty?

From logical agents to probabilistic reasoning

Acting Under Uncertainty

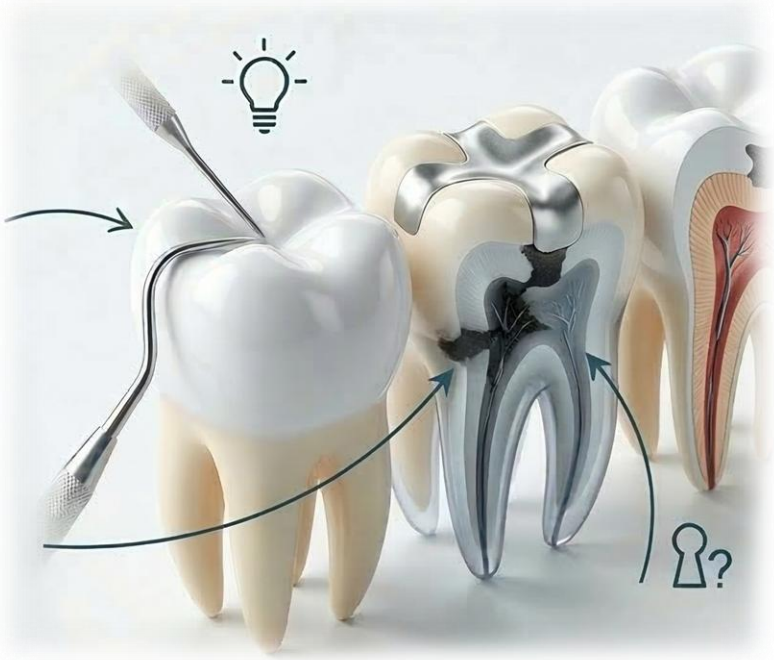
Agents in the real world can never be fully certain about what is happening. Why?

Partial Observability

The agent cannot see the full state of the world.

A dentist’s probe can detect surface issues, but cannot see decay hidden between teeth or beneath old fillings.

Critical information remains out of reach, forcing the agent to act on incomplete knowledge.

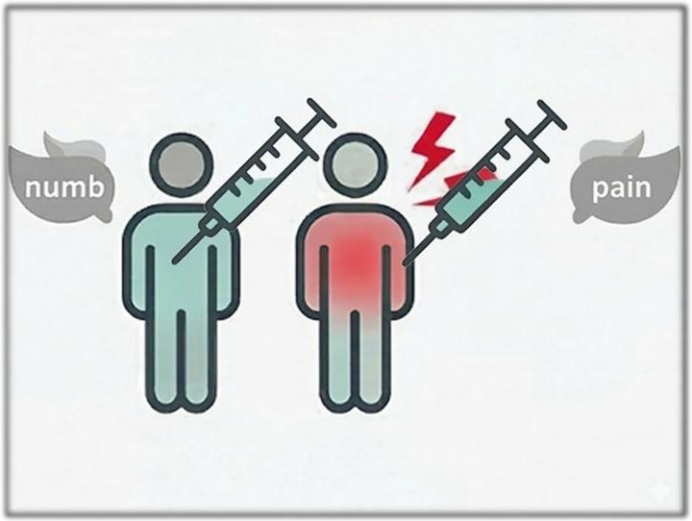


Nondeterminism

The same action can lead to different outcomes.

Injecting the same dose of local anaesthetic may fully numb one patient but leave another still feeling pain, depending on nerve anatomy.

Even with identical inputs, the world responds differently — the agent cannot predict with certainty what will happen next.

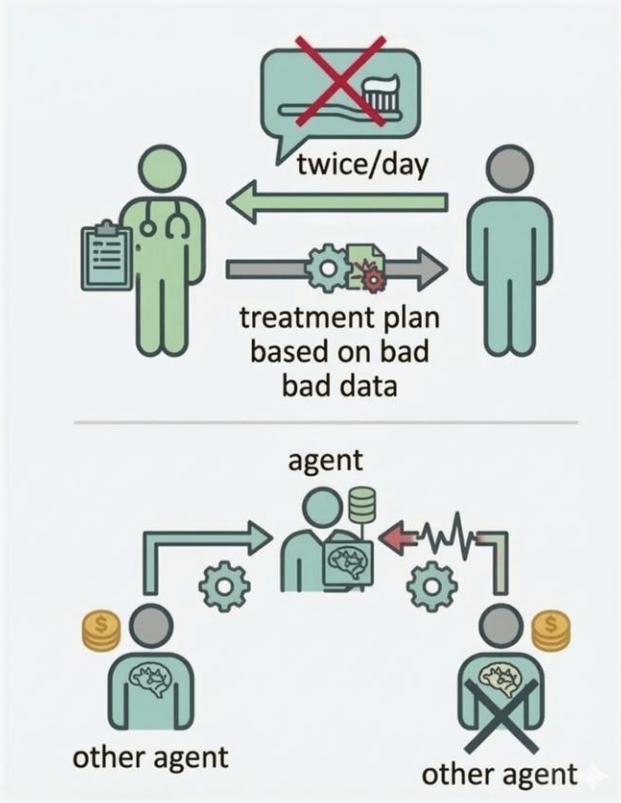


Adversaries

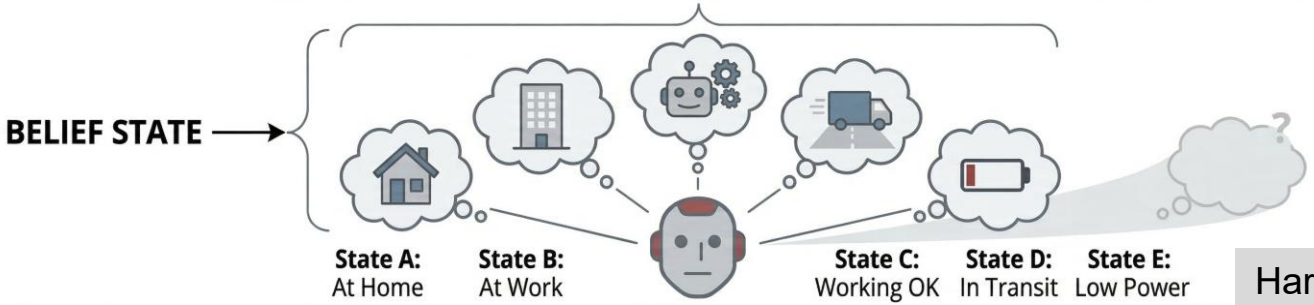
Other agents influence outcomes in unpredictable ways.

A patient claims they brush twice a day, but actually doesn't. The dentist's treatment plan is based on false information.

Other agents act in their own interest, providing misleading data that corrupts the agent's model of the world.



How do agents cope today?



Hard to track all possibilities!

They maintain a **belief state** — a representation of all possible world states the agent might be in. But this approach has serious limitations, as we'll see next.

Problems with Purely Logical Approaches

1

Too many possibilities

The agent must consider every possible explanation for what it observes — even extremely unlikely ones. The belief state becomes huge.

2

Huge contingency plans

A correct plan must handle every eventuality. As we add more contingencies, the plan grows arbitrarily large and becomes unusable.

3

No guaranteed solution

Sometimes no plan can guarantee success. Yet the agent must still act — it needs a way to compare imperfect plans.

Key takeaway: Pure logic demands certainty. In the real world, certainty is unattainable.

The Qualification Problem

Example: Dental Treatment Plan

A dentist must decide whether to fill a cavity in a patient's tooth. They create **Plan A**: drill and fill immediately.

Can the logical agent ***prove* this plan will work? No!**

What logic must guarantee:

"The plan will succeed, as long as...

- ✗ the drill doesn't slip
- ✗ the patient doesn't move
- ✗ the anaesthetic works
- ✗ the filling material bonds correctly
- ✗ and ... "

The list of conditions is infinite!

The qualification problem: Every logical rule needs an almost infinite list of exceptions to be truly correct.

Uncertainty in Logical Rules

Example: Medical Diagnosis

A simple logical rule:

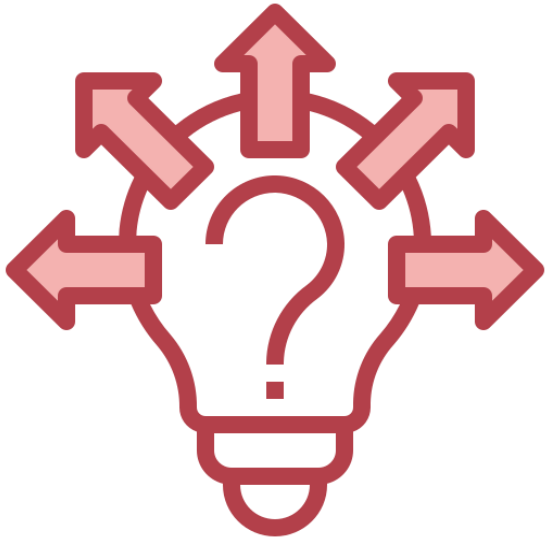
```
toothache → cavity
```

But this is wrong! A toothache can also come from gum disease, an abscess, or other problems.

Reversing the rule also fails:

```
cavity → toothache
```

Not all cavities cause pain!



Degrees of Belief

Instead of true/false, we assign probabilities — numbers expressing how strongly we believe something is true.

Numbers between 0 (certainly false) and 1 (certainly true). This is probability theory.

Uncertainty and Rational Decisions

Probabilities alone are not enough. Which treatment plan should the dentist choose?

Plan	Description	P(pain-free)	Problem
T_Q	**Quick fill (30 min)**	**85%**	**Risk of incomplete treatment**
T_R	Root canal (2 hrs)	97%	Long, uncomfortable procedure
T_E	Full extraction (10 min)	≈ 100%	Permanent tooth loss

The problem:

Higher probability ≠ better decision. T_E has the highest probability of being pain-free, but the cost (permanent tooth loss) is unacceptable.

The dentist needs to balance:
probability of success, cost of outcomes, personal preferences

Utility Theory

Core Idea

Every outcome has a degree of usefulness (**utility**) to an agent. The agent prefers states with higher utility.

Utility is **subjective and agent-relative** — there is no universal standard.

Example: Anesthetic Injection

Same anaesthetic injection — two patients:

Patient A

U(needle) = **0.8**

"Terrified of pain – happy to be numb."

Patient B

U(needle) = **0.2**

"Terrified of needles – rather feel the pain."

0 = worst outcome, 1 = best outcome — each agent's scale is personal and cannot be compared to others.

Decision Theory

Decision Theory = Probability Theory combined with Utility Theory

Principle of Maximum Expected Utility (MEU)

A rational agent chooses the action with the **highest expected utility**, calculated as:

$$\text{Expected Utility} = \Sigma \; (P(\text{outcome}) \times U(\text{outcome}))$$

Applied to the Dentist:

Plan	P(pain-free)	U(pain-free)	P(fail)	U(fail)	Wait cost	EU ≈
T_Q	**0.85**	**80**	**0.15**	**–10**	**–5**	**★ 61.5**
T_R	0.97	70	0.03	–10	–20	47.6
T_E	1.0	40	0.0	–10	–5	35.0

The Decision-Theoretic Agent

How does an agent put all of this together? Here are the 5 steps:

1

Maintain belief state

Probabilities of world states

2

Update beliefs

Incorporate new percepts

3

Predict outcomes

Estimate results for each action

4

Compute exp. utility

$\Sigma\ P(o) \times U(o)$ for each action

5

Choose best action

Action with highest EU

vs. Logical Agents: Logical agents track which states are *possible*. Decision-theoretic agents track how *probable* each state is — and what matters to the agent.

PART 2

The Language of Probability

Foundations, notation, posterior probability and full joint probability distribution

Basic Probability Notation

Random variable

A function from outcomes to values

Weather ∈ {sunny, rainy, cloudy, snowy}
Cavity ∈ {cavity, ¬cavity}

Sample space

Ω — the set of all possible worlds (outcomes)

Ω = {(sunny ∧ cavity), (sunny ∧ ¬cavity),
(rainy ∧ cavity)...}

Prior vs. Posterior Probability

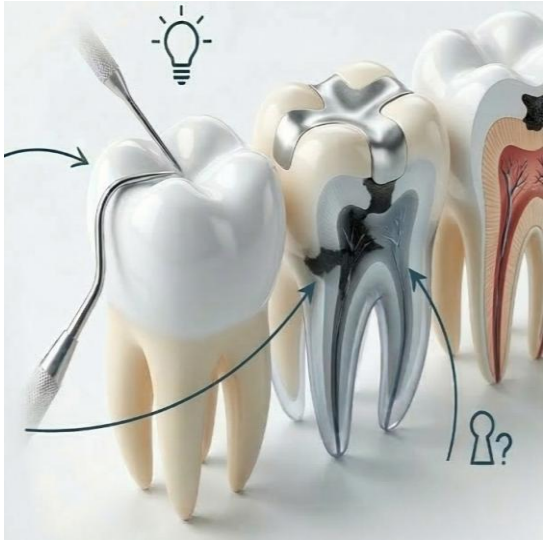
Knowledge changes – probabilities change

Prior Probability

P(cavity)

= 0.1

Belief before any evidence



Posterior Probability

P(cavity | toothache)

= 0.47

Updated on toothache evidence

Posterior Probability

Conditional probability

e.g. P(cavity | toothache)

$$P(A|B) = P(A \cap B) / P(B)$$

Probability of A, given B is already true.

Product rule

e.g. P(cavity∩toothache)

$$P(A \cap B) = P(B) \times P(A|B)$$

Rearrangement of the conditional probability definition.

Full Joint Probability Distribution

	Toothache	Catch	Cavity	P
ω1	**true**	**true**	**true**	**0.072**
ω2	**true**	false	**true**	0.008
ω3	**true**	**true**	false	0.0045
ω4	**true**	false	false	0.0855
ω5	false	**true**	**true**	**0.018**
ω6	false	false	**true**	0.002
ω7	false	**true**	false	0.0405
ω8	false	false	false	**0.7695**

Q1: What is P(Cavity = true)?

P(Cavity = true) = 0.072 + 0.008 + 0.018 + 0.002 = ***0.1***

Q2: P(Cavity=true | Toothache=true) = ?

P(Cavity = true | Toothache=true)
= P(Cavity = true ∧ Toothache=true) / P (Toothache=true)
= (0.072+0.008)/0.17 = **0.47**

Axioms of Probability

	Toothache	Catch	Cavity	P
ω1	**true**	**true**	**true**	**0.072**
ω2	**true**	false	**true**	0.008
ω3	**true**	**true**	false	0.0045
ω4	**true**	false	false	0.0855
ω5	false	**true**	**true**	**0.018**
ω6	false	false	**true**	0.002
ω7	false	**true**	false	0.0405
ω8	false	false	false	**0.7695**

1. **Probability bounds:** every probability is between 0 and 1
2. **Total probability (Normalization):** all probabilities in the distribution sum to 1
3. **Inclusion-exclusion:** $P(a \vee b) = P(a) + P(b) - P(a \wedge b)$

The Problem with Full Joint Distributions

	Toothache	Catch	Cavity	P
ω1	**true**	**true**	**true**	**0.072**
ω2	**true**	false	**true**	0.008
ω3	**true**	**true**	false	0.0045
ω4	**true**	false	false	0.0855
ω5	false	**true**	**true**	**0.018**
ω6	false	false	**true**	0.002
ω7	false	**true**	false	0.0405
ω8	false	false	false	**0.7695**

$2 \times 2 \times 2 = 2^n$ entries. With 100 variables: $2^{100} \approx 10^{30}$

We need a compact representation

- Computationally too large to store
- Not enough storage to hold the full table
- Not enough data to estimate all parameters reliably

Bayes' Rule

Product rule

e.g. P(cavity∩toothache)

$$P(a \cap b) = P(b) \times P(a|b)$$

$$P(a \cap b) = P(a) \times P(b|a)$$

Rearrangement of the conditional probability definition.

$$P(b) \times P(a|b) = P(a) \times P(b|a)$$

$$P(a|b) = P(a) \times P(b|a) / P(b)$$

PART 3

Independence & Naïve Bayes

Making probabilistic inference tractable

Recap: Bayes' Rule

Bayes' Rule

$$P(H \mid e) = P(e \mid H) \cdot P(H) \;/\; P(e)$$

posterior	likelihood	prior	evidence
what we want	model of evidence	background belief	normalizer

Real-world analogy: Bayes' Rule is like a translator. Doctors know how diseases cause symptoms (causal direction). But they need to know: given a symptom, how likely is the disease? (diagnostic direction). Bayes' Rule translates between them.

Bayes' Rule: Flipping the Direction

The dentist knows causes → symptoms. But needs symptoms → causes.

What the dentist knows (causal)

`P(toothache | cavity) = 0.8`

80% of cavities cause toothache.

This is based on medical knowledge — how diseases work.

What the dentist needs (diagnostic)

`P(cavity | toothache) = ???`

Patient has toothache — is it a cavity?

This is the diagnostic question. Bayes' Rule does the flip!

$$P(cavity|toothache) = P(toothache|cavity) \cdot P(cavity) / P(toothache)$$

Causal knowledge (cause → symptom) is stable and universal. Diagnostic knowledge (symptom → cause) depends on context.

Why Causal Knowledge is Robust

Causal facts don't change when the world changes. Diagnostic facts do.

Example: Meningitis causes a stiff neck in 70% of cases. P(stiff neck | meningitis) = 0.7

Normal times

`P(meningitis) = 1 in 50,000`

`P(stiff neck | meningitis) = 0.7`

Stiff neck? Probably just slept wrong.

During an epidemic

`P(meningitis) = 1 in 100`

`P(stiff neck | meningitis) = 0.7`

Stiff neck? Much more worrying now!

The causal probability (0.7) never changed. The disease works the same way.

But the diagnostic probability P(meningitis | stiff neck) changed dramatically.

Store causal knowledge + use Bayes' Rule to adapt. Don't hard-code diagnostic probabilities — they break when the world changes.

Normalization: The Practical Trick

Computing P(toothache) in the denominator is annoying. Here's a shortcut.

$$P(\text{Cavity} \mid \text{toothache}) = \alpha \cdot P(\text{toothache} \mid \text{Cavity}) \cdot P(\text{Cavity})$$

α just makes the probabilities sum to 1. Compute the numerator for each case, then normalize.

1. Cavity = true: $P(\text{toothache}|\text{cavity}) \cdot P(\text{cavity}) = 0.8 \times 0.1 = 0.08$
2. Cavity = false: $P(\text{toothache}|\neg\text{cavity}) \cdot P(\neg\text{cavity}) = 0.1 \times 0.9 = 0.09$
3. Normalize: $0.08 / (0.08 + 0.09) = 0.08 / 0.17 \approx 47\%$

Even with a toothache, only 47% chance it's a cavity! Toothaches have many causes. Can we do better with more evidence?

Two Symptoms Are Better Than One

Adding the probe catch: P(cavity | toothache AND catch)?

Given information

```
P(toothache | cavity)  = 0.8
P(catch | cavity)      = 0.9
P(toothache | ¬cavity) = 0.1
P(catch | ¬cavity)     = 0.05
P(cavity) = 0.1
```

The calculation

Cavity = true:

`0.8 × 0.9 × 0.1 = 0.072`

Cavity = false:

`0.1 × 0.05 × 0.9 = 0.0045`

Normalize:

`0.072 / 0.0765 ≈ 94%!`

Toothache alone: P(cavity) ≈ 47%

Toothache + Catch: P(cavity) ≈ 94%

Two independent clues confirming the same diagnosis make it far more certain. This is the power of combining evidence.

When Things Don’t Matter

Not everything is connected — some variables are truly independent

The question

Does today’s weather affect whether you have a cavity?

Obviously not!

The payoff

One big table → two small tables:

`32 entries → 8 + 4 = 12`

62.5% reduction in storage!

But here’s the tricky part…

Toothache and Catch ARE related — if the probe catches, it’s more likely you have a cavity, which means more toothaches.

Full independence is rare in the real world. We need a more subtle idea…

$P(A,B) = P(A) \cdot P(B)$ only when A and B are truly independent. Most real variables have some connection.

Conditional Independence: The Key

Once you know the cause, the effects become independent

Before knowing about the cavity

Toothache and Catch are connected:

Probe catches → likely cavity → likely toothache

They tell you about each other.

After learning cavity status

Toothache and Catch are unrelated:

Toothache = nerve response.

Catch = probe contact.

Separate mechanisms, no link.

$$P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) \cdot P(\text{Catch} \mid \text{Cavity})$$

This is conditional independence: given the cause, effects become independent of each other.

The cause “separates” its effects. This is one of the most important ideas in AI — it makes everything tractable.

What This Lets Us Do

Replace one huge table with many small, simple pieces

1. **How common are cavities?**
   `P(Cavity) = 0.1 → just 1 number`

2. **How often does a cavity cause toothache?**
   `P(Toothache | Cavity) → 2 numbers`

3. **How often does the probe catch?**
   `P(Catch | Cavity) → 2 numbers`

The payoff: only 2n+1 numbers needed

Original table: 7 numbers

Decomposed: only 5 numbers

With 3 variables: small saving.
With 10 variables: 1,023 → 21
With 100 symptoms:
10^{30} → just 201!

Conditional independence turns exponential problems into linear ones. This is the foundation of Naïve Bayes.

The Naïve Bayes Model

Assume every effect is independent given the cause, then just multiply

$$P(\text{Cause} \mid e_1, e_2, \ldots, e_n) = \alpha \cdot P(\text{Cause}) \cdot P(e_1|\text{Cause}) \cdot P(e_2|\text{Cause}) \cdot \ldots \cdot P(e_n|\text{Cause})$$

Why "naïve"?

We assume all symptoms are independent given the disease.

That's rarely perfectly true!

But it works surprisingly well — even if the exact probabilities are off, the ranking of diagnoses stays correct.

Complexity reduction

Without independence:

$O(2^n)$ parameters

With Naïve Bayes:

$O(n)$ parameters

Fast, scalable, practical!

Naïve Bayes: one cause produces many effects. Multiply the prior by each symptom's contribution. Simple, fast, and it works.

Where You See This in the Real World

Same pattern everywhere: one cause, many observable effects

Spam filtering

Cause: spam or not

Effects: which words appear in the email

Each word contributes independently to the spam score.

Medical diagnosis

Cause: disease

Effects: symptoms, test results, patient history

Exactly our cavity example, but with dozens of symptoms.

Language detection

Cause: which language

Effects: character patterns and common words

Each pattern independently points to a language.

Naïve Bayes is one of the most widely used classifiers in AI. Simple, fast, interpretable, and surprisingly accurate.

What We've Learned

1. **The full joint table can answer any question — but it doesn't scale**
   With 100 variables, the table has more entries than atoms in the universe.

2. **Bayes' Rule flips causal knowledge into diagnostic answers**
   Store stable causal facts, adapt to new evidence via normalization.

3. **One symptom = 47%, two symptoms = 94%**
   Combining independent evidence makes diagnosis dramatically more certain.

4. **Conditional independence is the key to tractability**
   Once you know the cause, effects separate. Exponential → linear.

These ideas lead to Bayesian Networks (Ch. 13) — a powerful framework for probabilistic reasoning at scale.

Looking Ahead: Where Does This Lead?

Ch. 13

Probabilistic Reasoning

Bayesian networks — compact representations of joint distributions using conditional independence

Ch. 14

Reasoning Over Time

Hidden Markov models, Kalman filters — extending probabilistic reasoning to sequences and dynamics

Thank You

Questions & Discussion

Chapter 12: Quantifying Uncertainty
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